# PUNJAB
# RETROFITTING SCHOOLS AND HEALTH FACILITIES

**Sector:** Infrastructure & Urban Resilience
**Hazard adressed:** Flood
**Targeted assets:** Education facilities, Health facilities


**Description:** This measure involves structural retrofitting of public buildings – specifically schools and colleges – located near river banks across 9 Punjab districts. Interventions include improving site drainage and constructing flood barriers around qualifying buildings.

**Priority Districts:** Lahore, Sheikhupura, Nankana Sahib, Jhang, Toba Tek Singh, Multan, Muzaffargarh, Bahawalpur, Rajanpur.

## TECHNICAL SPECIFICATIONS/DIMENSION

**Unit of implementation:** Per building.

**Scale assumptions:** Educational buildings (schools and colleges) within 250m of river lines in target districts.

**Key design parameters:** Improving site drainage; construction of flood barriers; eligibility criterion = within 250m of river lines.

**Time horizon:** 2026–2050.

## IMPACT

**Risk reduction level:** High Effectiveness – significant reduction (Expert Rank 2). Damages are reduced by 90% corresponding to 1-in-100-year flood events.

**Time to effectiveness:** Medium-term; effectiveness realized progressively as individual buildings are strengthened over three implementation phases (2026–2040).

**Expected impact:** Expected to significantly reduce structural flood damage to educational buildings located in close proximity of rivers through flood barriers and drainage improvement.

## IMPLEMENTATION CONSIDERATIONS

**Enabling conditions:** Structural engineering assessments for each building; hydraulic design for flood barriers; asset surveys to identify eligible buildings; inter-departmental coordination between Education and Irrigation departments.

**Required institutions:** Punjab School Education Department; Punjab Higher Education Department; Punjab Irrigation Department (for river proximity mapping); PDMA Punjab.

**Barriers:** Asset survey and eligibility mapping complexity; procurement of specialized contractors; coordination across multiple departments.

## COST ASSUMPTIONS

**Unit cost:** Not a single unit cost used, cost varies by building size.

**Capital Expenditure (CAPEX):**
For Urban Resilience: USD 0.9 million per year (2026 – 2030). For Infrastructure: average USD 2.2 million per year (2026 – 2040).

**Operational Expenditure (OPEX):**
For Urban Resilience: USD 0.06 million per year (2031 – 2050). For Infrastructure: USD 0.5 million per year (2040 – 2050).

## CO-BENEFITS & TRADE-OFFS

**Environmental:** Improved site drainage at school/college campuses may reduce localized waterlogging; flood barriers can help protect riparian green spaces.

**Social:** Protected school and college infrastructure ensures continuity of education services during flood events; reduces displacement and trauma for students; schools often serve as temporary flood shelters.

**Economic:** Avoided costs of flood damage and reconstruction of buildings; continuity of education services reduces longer-term human capital losses.

**Potential risks:** CAPEX assumes existing structures can be adequately protected; some buildings may require full reconstruction and/ or relocation rather than retrofitting.

## BENEFICIARIES

**Primary beneficiaries:** Students and staff in schools and colleges; communities using these facilities as flood shelters; Punjab Education departments.

**Distributional aspects:** Benefits accrue disproportionately to communities living near rivers.

Note: This factsheet is to be used as a rapid planning and appraisal tool. All costs and impacts should be validated through field assessments, stakeholder consultation and detailed feasibility studies.